

Tobias Wren

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Philadelphia, PA

Objective

Early-career behavioral geneticist with a strong foundation in quantitative genetics, twin study methodology, and neuropsychiatric trait analysis. Seeking a research or doctoral position where I can apply my experience in genomic data analysis and behavioral phenotyping to advance understanding of gene-environment interactions in complex traits.

Education

M.S. Behavioral Genetics

Temple University, 2023

B.S. Psychology

Villanova University, 2021

Skills

R, Python, PLINK, GCTA, SPSS, STATA, bash scripting, SNP array analysis, genome-wide association studies (GWAS), twin modeling (ACE/ADE), heritability estimation, behavioral phenotyping, statistical genetics, literature review, manuscript preparation, IRB protocols, REDCap, Zotero

Awards

Dean's List for Academic

Excellence, Temple University (2022, 2023)

Departmental Distinction —

Master's Thesis, Temple University (2023)

Psi Chi Honor Society

Induction, Villanova University (2021)

Travel Award — Behavior Genetics Association Annual Meeting (2023)

Experience

Junior Research Associate

, University of Pennsylvania

San Francisco, CA

Sep 2023 - Present

Contributes to an ongoing longitudinal twin study examining heritability of anxiety and executive function traits across adolescence, managing data collection, cleaning, and preliminary analysis for a cohort of 300+ twin pairs. Runs GWAS pipelines using PLINK and GCTA on institutional HPC cluster, handling quality control, population stratification correction, and results visualization.

Projects

Master's Thesis — Heritability of Risk-Taking Behavior in Young Adults (2023)

Classical twin study using 180 MZ/DZ twin pairs estimating genetic contributions to risk-taking behavior via BART. Fitted univariate ACE models in R (OpenMx); found $h^2 = 0.41$. Awarded departmental distinction.

Gene-Environment Interaction in Childhood Anxiety (2022)

Used ABCD study data ($n=4,500+$) to test whether polygenic risk scores for anxiety moderated the effect of adverse childhood experiences on internalizing symptoms. Linear mixed-effects modeling in R.

Polygenic Score Analysis — Educational Attainment and Cognitive Performance (2022)

Constructed and validated polygenic scores from Lee et al. (2018) GWAS summary statistics. Tested predictive validity in an independent sample, replicating published effect sizes and extending findings to an understudied subgroup.

Behavioral Phenotyping Protocol Development (2021)

Designed an impulse control assessment battery (BIS-11, Stop Signal Task, delay discounting) administered via REDCap. Piloted with 30 participants and authored IRB amendment for expanded recruitment.